

E. L. Lehmann, J. P. Romano:
Testing statistical hypotheses

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This monograph under review is the third edition, now jointly with Joseph P. Romano, of Erich L. Lehmann's classical graduate text on "Testing statistical hypotheses". The main material of the second edition from 1986 has comprehensively been reorganized and widely been extended, in particular with respect to a detailed discussion of the *asymptotic* optimality of testing procedures.

The book is divided into two major parts. Part I on "Small sample theory" covers more or less the topics from the second edition, i.e. the general decision problem, the probability background, uniformly most powerful (UMP) tests, unbiasedness, invariance, linear hypotheses, the minimax principle, and conditional inference. A new chapter on multiple testing and simultaneous inference has been added, including some recent optimality results, whereas some of the material on the multivariate linear hypotheses has been eliminated. Most of the asymptotic aspects from the second edition have now been moved to Part II on "Large-sample theory".

The latter part contains the most comprehensive addition to this third edition, covering basic large sample theory (e.g. asymptotic UMP tests, convergence concepts, robustness aspects, nonparametrics), quadratic mean differentiable families (contiguity, Le Cam's lemmas, likelihood methods), large sample optimality (testing sequences, metrics, ARE, AUMP tests, asymptotically normal experiments, applications to parametric and nonparametric models), testing goodness of fit (Kolmogorov–Smirnov, Cramér–von Mises, Anderson–Darling, Pearson's chi-squared test, Neyman's smooth tests, weighted quadratic tests, power), and general large sample methods (permutation and randomization tests, bootstrap, subsampling).

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The material of the book can certainly be taught in a two-semester graduate course on the testing of statistical hypotheses, divided in a natural way into the two major parts on small and large sample theory, respectively. A deep knowledge of general measure theory and integrals is not really required for the understanding, but will be helpful in following the proofs, in particular those of the asymptotic results. Some necessary preliminaries on equivalence relations and groups, convergence of functions and metric spaces, Banach and Hilbert spaces, dominated families of distributions, and the weak compactness theorem have been collected in an appendix. More than 300 new problems and many examples from parametric, but also from nonparametric statistics, will certainly be supportive in illustrating the general theory. Additional insight into the historical background and recent developments is given in some notes at the end of each chapter. More than 1,000 original references are provided.

In conclusion, this third edition of “Testing statistical hypotheses” is still an excellent and demanding treatment of modern statistical test theory. There is no doubt that it remains and will even more be used as a standard monograph by graduate students and researchers in the field.